

The minimum number of edges in a pancyclic graph

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Abstract

For $n \geq 3$, let $n + h(n)$ be the least number of edges in a graph on n vertices that contains a cycle of every length from 3 to n . Bondy stated in 1971 that $\log_2(n-1) - 1 \leq h(n) \leq \log_2 n + \log_* n + O(1)$. Erdős Problem #1016 asks whether the lower bound can increase to match the upper bound. We prove that $h(n) = \log_2 n + \log_* n + O(1)$.

The main estimate concerns a connected graph with large cycle rank. Remove a union of cycles with few edges and few components. Restrict a uniformly random even edge set to the remaining edges. The probability that this restriction is a forest is at most $1/2 + o(1)$. This estimate gives a near-halving recurrence for the normalized cycle-length capacity. Iteration gives the $\log_* n$ term.

A Lean 4 formalization verifies the proof.

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1 Introduction

A graph on n vertices is *pancyclic* if it contains a cycle of every length $3, 4, \dots, n$. Let $n + h(n)$ be the least number of edges of a pancyclic simple graph on n vertices, and let $\log_* x$ be the least $j \geq 0$ such that j applications of $\log_2$ take x to a value at most one.

Theorem 1.1. *There is an absolute constant A such that, for every $n \geq 3$,*

$$\log_2 n + \log_* n - A \leq h(n) \leq \lfloor \log_2 n \rfloor + \log_* n + 1.$$

In particular, $h(n) = \log_2 n + \log_ n + O(1)$.*

Bondy [4] stated without proof that $\log_2(n-1) - 1 \leq h(n) \leq \log_2 n + \log_* n + O(1)$. A cycle count gives the lower bound. A connected graph with $n + h$ edges has at most $2^{h+1} - 1$ cycles. A pancyclic graph needs $n - 2$ of these cycles.

Griffin [7] gave a written proof and computed small values. Sridharan [10] and George, Khodkar and Wallis [6] give constructions. Alon and Krivelevich [1] give constructions for random graphs. Can the lower bound increase to $\log_2 n + \log_* n - O(1)$? Does even $h(n) - \log_2 n \rightarrow \infty$ hold? These are the questions in Erdős Problem #1016 [3].

Theorem 1.1 answers both questions.

1.1 The main estimate and an outline

A cycle count does not use the requirement that the lengths fill an interval. The following estimate supplies the additional constraint. For a graph G , let $\mathcal{Z}(G)$ be its binary cycle space, the even edge sets of G , of dimension $\beta(G) = |E(G)| - |V(G)| + \text{comp}(G)$. Let ξ be uniform on $\mathcal{Z}(G)$, and for an edge set M put

$$P_G(M) = \mathbb{P}(\xi \cap M \text{ is a forest}).$$

Theorem 1.2 (Forest estimate). *There is r_0 such that the following holds for all $r \geq r_0$. Let G be a connected simple graph with $\beta(G) = r$, and let $W \subseteq G$ be a nonempty union of cycles with $m = |E(W)| \leq \sqrt{r}$ edges and $c = \text{comp}(W) \leq (\log_2 r)/100$ components. Then*

$$P_G(E(G) \setminus E(W)) \leq \frac{1}{2} + \frac{20}{\log^{(3)} r}. \quad (1.1)$$

Many parts of a graph can share one parity bit. That bit can make all these parts acyclic with probability $1/2$. This explains the threshold. Section 2 gives the zero-cut correlation ratio for two regions: $2^{\ell-1}$. Here ℓ counts the links between the regions. The section also shows that many disjoint small cyclic regions reduce the forest probability to approximately $1/2$.

Section 3 proves Theorem 1.2. We reduce the maximum degree to three and keep W as a marked set with few components. We then delete short cycles. The nonbacktracking spectrum supplies many moderately long cycles. We apply the second moment method to the events that such a cycle is a component of ξ .

Equation (3.6) gives the main link bound. At most $c + o(\log r)$ components link two short cycles. Thus the witness components control all correlations. The proof needs no expansion assumption.

Section 4 uses Theorem 1.2 to prove the recurrence

$$\Phi(2^{256R}) \leq \left(\frac{1}{2} + \frac{20}{\log^{(2)} R} \right) \Phi(R) + 2^{2-R}$$

for the normalized cycle-length capacity Φ . Each step increases the rank by one exponential level and reduces Φ by approximately one half. The errors are summable. Thus $\Phi(r) = O(2^{-\log_* r})$, which gives the lower bound. Section 5 gives the construction.

1.2 Formal verification

The repository <https://github.com/JWKNT/erdos1016> contains the proof formalization in Lean 4 [5] with Mathlib [9]. This paper describes commit `a38913e`. It has 40,923 lines in 247 active modules, including the entry point. The build uses Lean v4.19.0 and a pinned Mathlib revision.

This commit passed fresh verification in the repository’s continuous integration. The check rebuilds every active project module from source. It also audits the statements and axioms. GitHub Actions run `36197260250` records the result.

In the `Erdos1016` namespace, `mainTheorem` proves the two-sided asymptotic assertion in Theorem 1.1. Its additive constants are unspecified. The declaration `Problem1016.upper_bound_integer` proves the explicit upper bound. Here $h(n)$ is the minimum of $|E(G)| - n$ over pancyclic simple graphs G on the vertex type `Fin n`, and $\log_* n$ is the least k with $n \leq T(k)$, where $T(0) = 1$ and $T(k+1) = 2^{T(k)}$.

Theorem 1.2 corresponds to `ShortProof.fewComponentForestEstimate`, with the same hypotheses and bound. These theorems depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`. Appendix A lists the Lean declarations for the main steps.

1.3 Conventions

All graphs are finite. Every feasible prescribed-boundary class of edge sets has $2^{\beta(G)}$ members. Regions are induced, $P_G(H) = P_G(E(H))$, and a region is *cyclic* if it contains a cycle. For a region A , δA is its cut, and “ $\delta A = 0$ ” is the event that ξ avoids it. Auxiliary graphs may have loops and parallel edges, with loops counted twice in degrees. All logs are binary except $\ln$; $\log^{(j)} t$ means j successive binary logarithms, and $\log_*$ stops at one.

2 Cut correlations

For a nonnegative random variable Z with $\mu = \mathbb{E}Z > 0$ and a constant $K \geq 0$, Cauchy–Schwarz gives

$$\mathbb{E}Z^2 \leq 2\mu^2 + K\mu \implies \mathbb{P}(Z = 0) \leq 1 - \frac{\mu^2}{\mathbb{E}Z^2} \leq \frac{1}{2} + \frac{K}{4\mu + 2K}. \quad (2.1)$$

Lemma 2.1 (Exceptional cuts). *In a connected graph, let A be a connected region with cut size d . Among connected regions B disjoint from A whose zero-cut events satisfy*

$$\mathbb{P}(\delta A = \delta B = 0) > 2\mathbb{P}(\delta A = 0)\mathbb{P}(\delta B = 0),$$

no vertex-disjoint subfamily has more than d members.

Proof. Contract A and a disjoint subfamily of the B ’s to vertices. The cycle-space projection is onto. Thus it preserves their joint cut law. For two vertices v, w , count as *links* the vw edges and the components of $G - \{v, w\}$ attached to both. If there are ℓ links, rank subtraction gives

$$\mathbb{P}(\delta v = \delta w = 0) = 2^{\ell-1} \mathbb{P}(\delta v = 0) \mathbb{P}(\delta w = 0). \quad (2.2)$$

Fix v . Append its d incidences as terminal leaves to a spanning forest of $G - v$. Prune nonterminal leaves. Each link to w gives a branch at w that contains a terminal. Thus each vertex with at least three links is a branch vertex of the pruned forest. This forest has at most d branch vertices. $\square$

Corollary 2.2 (Small cuts). *For $k \geq 1$ vertex-disjoint connected cyclic regions $H_1, \dots, H_k$ with cuts at most d in a connected graph G ,*

$$P_G\left(\bigcup_i E(H_i)\right) \leq \frac{1}{2} + \frac{(d+1)2^d}{2k}. \quad (2.3)$$

Proof. Let E_i say that the i th cut is zero and its inner even set is nonempty. Its probability p_i is at least 2^{-d-1} . Given zero cuts, the inner even sets are independent. Thus the pairwise correlation ratio equals the zero-cut ratio. By Lemma 2.1, each event has at most d exceptional partners in this disjoint family.

For the other pairs, use $\mathbb{P}(E_i E_j) \leq 2p_i p_j$. For exceptional partners, use $\mathbb{P}(E_i E_j) \leq p_i$. Thus $Z = \sum_i \mathbf{1}_{E_i}$ satisfies

$$\mu \geq k2^{-d-1}, \quad \mathbb{E}Z^2 \leq 2\mu^2 + (d+1)\mu.$$

Each E_i excludes a forest. Thus (2.1) proves the claim. $\square$

3 Proof of the forest estimate

Let G and W satisfy the hypotheses of Theorem 1.2, with r large. Set

$$x = \log r, \quad y = \log x, \quad z = \log y, \quad D = \lfloor x/10 \rfloor, \quad L = \lfloor x\sqrt{y} \rfloor,$$

and suppose the probability in (1.1) exceeds $\frac{1}{2} + \varepsilon$, where $\varepsilon = 20/z$. We obtain a contradiction. All estimates below are uniform under the two witness bounds.

3.1 Preserve witness components during degree reduction

Replace each vertex of degree $d > 3$ by a path of $d - 2$ cubic vertices. Put two old incidences at each end. Put one old incidence at each internal vertex. If $k \geq 2$ incidences belong to W , put these incidences first along the path. Mark the initial segment that meets them. This segment is connected and has at most k vertices.

Also mark unchanged witness vertices. In the expanded graph G' , the witness lifts to a subgraph on a marked set P with

$$|P| \leq 2m, \quad \text{comp}(G'[P]) \leq c.$$

Contraction of the splitting trees gives a cycle-space bijection. Lift an original forest to the expanded graph. Add all tree edges to this lift. The result is still a forest. Thus restriction to the complement of P can only increase the forest probability.

Prune unmarked leaves. Suppress unmarked degree-two vertices. Their even coordinates are zero at leaves and equal along suppressed paths, respectively. These operations preserve the cycle-space law and the forest event outside P . They remove no marked vertex. Suppression can join marked components, but cannot split them.

Call the resulting connected multigraph Γ . It has rank r and maximum degree three. Every unmarked vertex is cubic. The following bounds hold:

$$|P| \leq 2\sqrt{r}, \quad \text{comp}(\Gamma[P]) \leq c, \quad |\Gamma| = 2r + O(\sqrt{r}), \quad P_\Gamma(\Gamma - P) > \frac{1}{2} + \varepsilon. \quad (3.1)$$

The order identity follows from $|\Gamma| = 2r - 2 + \sum_v (3 - \deg_\Gamma v)$. Only marked vertices contribute to this sum.

3.2 Delete short cycles without connections between them

Suppose $\Gamma - P$ contains $\lceil \sqrt{r} \rceil$ disjoint cycles of length at most D , including loops and digons. Their induced regions have cuts at most D . Then (2.3) gives $P_\Gamma(\Gamma - P) \leq \frac{1}{2} + O(xr^{-2/5})$. This is a contradiction.

Otherwise, choose a maximal such packing. Let S be the union of its vertex sets. Then $|S| \leq D\sqrt{r}$, and

$$J = \Gamma - (P \cup S)$$

is simple with girth greater than D , order $v = (2 + o(1))r$, and deficit

$$d_J := \sum_{u \in J} (3 - \deg_J u) \leq 3(|P| + |S|) = O(\sqrt{r} \log r). \quad (3.2)$$

The proof does not require P or $P \cup S$ to be connected.

3.3 Supply cycle weight

Let B , A_0 and Δ be the directed-edge nonbacktracking, adjacency and degree matrices of J , respectively. Set $M(t) = t^2 I - tA_0 + \Delta - I$. Compare this with the Ihara–Bass identity [2, 8]. Then

$$M(2) = 2(\Delta - A_0) + 3I - \Delta \succeq 0, \quad \mathbf{1}^T M(2 - d_J/v) \mathbf{1} = 0.$$

The least eigenvalue of the real symmetric matrix $M(t)$ is continuous. It is nonpositive at $2 - d_J/v$ and nonnegative at 2. Thus $M(t)$ is singular for some $t \in [2 - d_J/v, 2]$, with $t > 1$. A null vector u gives the nonzero B -eigenvector $f(i, j) = tu_j - u_i$.

Conversely, a nonreal eigenvalue t of B gives $M(t)u = 0$ with $u \neq 0$. Multiplication by u^* gives a real quadratic with constant-to-leading ratio at most two. Thus $|t| \leq \sqrt{2}$. Even powers of real eigenvalues are nonnegative. Therefore

$$2^{-\ell} \operatorname{tr} B^\ell \geq (1 - d_J/(2v))^\ell - 3v 2^{-\ell/2} \quad (\ell \text{ even}).$$

By (3.2), the sum over even $6x \leq \ell \leq L$ gives

$$T := \sum_{\ell \leq L} \frac{\operatorname{tr} B^\ell}{2^\ell 2^\ell} \geq (\tfrac{1}{4} + o(1)) \ln \frac{L}{6x} = (\tfrac{1}{8} + o(1)) \ln y.$$

Set $s = \lfloor (D - 1)/2 \rfloor$. The girth bound makes an s -step nonbacktracking path between fixed endpoints unique. Let $N_t(u, w)$ count ordinary nonbacktracking walks of length t from u to w . Then

$$2^{-t} N_t(u, w) \leq \tfrac{3}{2} 2^{-s} \quad (t > s). \quad (3.3)$$

Let $V = \sum_{C \subset J, |C| \leq L} 2^{-|C|}$, over simple cycles. A nonsimple cyclically nonbacktracking walk contains a proper simple cycle. Its complementary ordinary closed nonbacktracking walk has length greater than D . Encode the walk by the cycle, an oriented start, the complement and the original root. The factor $1/(2\ell)$ cancels orientations and root positions. The start vertex and complementary length contribute at most L^2 .

Thus

$$0 \leq T - V \leq \tfrac{3}{2} L^2 2^{-s} V, \quad V \geq (\tfrac{1}{8} + o(1)) \ln y. \quad (3.4)$$

3.4 Normalize cycle indicators

Consider each cycle $C \subset J$ of length at most L . Let I_C be the event that C is a component of the random even set. Set

$$a_C = 2^{-|C|}, \quad p_C = \mathbb{P}(I_C), \quad X_C = (a_C/p_C)\mathbf{1}_{I_C}, \quad Z = \sum_C X_C.$$

Cubic ambient degree makes I_C exactly the event that the even set contains all cycle edges. These binary linear conditions are feasible. Thus $p_C \geq a_C$, $0 \leq X_C \leq 1$ and $\mathbb{E}X_C = a_C$.

For disjoint cycles C, C' , let $\ell(C, C')$ count their direct joining edges and the components of $\Gamma - (V(C) \cup V(C'))$ attached to both. Prescribed inner cycle words multiply zero-cut probabilities by separate inner factors. These factors cancel in the ratio (2.2). Thus

$$\mathbb{E}(X_C X_{C'}) = a_C a_{C'} 2^{\ell(C, C') - 1}. \quad (3.5)$$

3.5 The deleted cycles count as cyclic components

By (3.1), at most c components of $\Gamma - (V(C) \cup V(C'))$ meet P . Every other component F_i lies in $\Gamma - P$. Its cut d_i ends on the two cycles, and $\sum_i d_i \leq 2L$. Set $t_0 = \lfloor y/2 \rfloor$. At most $2L/t_0$ components have $d_i > t_0$. Among the remaining components, at most $O(t_0 2^{t_0}/\varepsilon)$ are cyclic, by (2.3) and (3.1).

If F_i is a tree, it cannot meet S . Every packed cycle that meets F_i lies wholly in it, because the packed cycle is disjoint from C, C' . Thus each such tree lies in the high-girth graph J . Cubic ambient degree gives $|F_i| = d_i - 2$.

If the tree meets both cycles and $d_i \leq t_0$, choose a path through it that joins the cycles. Choose this path with length at most t_0 . Include direct joining edges as paths of length one. These paths have disjoint interiors and distinct endpoints. Partition each cycle into consecutive vertex blocks of size at most $\lfloor D/4 \rfloor$.

Two paths between the same block pair would form a cycle of length at most $2t_0 + D/2 < D$. Thus there are only $O((L/D)^2)$ such paths. Uniformly in the pair,

$$\ell(C, C') \leq c + O\left((L/D)^2 + L/t_0 + t_0 2^{t_0}/\varepsilon\right) = c + o(x). \quad (3.6)$$

3.6 Complete the second moment bound

Call a disjoint pair exceptional when $\ell(C, C') \geq 3$. By Lemma 2.1, a disjoint packing of exceptional partners of C has at most $|\delta C| \leq L$ members. A maximal packing has union U_C of size at most L^2 . This union meets every exceptional partner. Equation (3.3) bounds the cycle weight through a fixed vertex by $O(L2^{-s})$. Thus (3.5) and (3.6) give

$$\sum_{C' \text{ exceptional to } C} \mathbb{E}(X_C X_{C'}) \leq a_C 2^{c+o(x)} O(L^3 2^{-s}) = o(1) a_C, \quad (3.7)$$

because $c \leq x/100$ and $s = x/20 + O(1)$.

Intersecting distinct cycles are incompatible. Nonexceptional disjoint pairs contribute at most $2a_C a_{C'}$. Also, $\mathbb{E}X_C^2 \leq a_C$. Thus, with $\mu = \mathbb{E}Z = V$,

$$\mathbb{E}Z^2 \leq 2\mu^2 + (1 + o(1))\mu.$$

Since every C lies in $\Gamma - P$, equations (2.1) and (3.4) give

$$P_\Gamma(\Gamma - P) \leq \mathbb{P}(Z = 0) \leq \frac{1}{2} + \frac{2 + o(1)}{\ln y} < \frac{1}{2} + \frac{20}{z},$$

which contradicts (3.1). This proves Theorem 1.2. $\square$

4 Cycle lengths and the lower bound

Let $\Phi(R)$ be the supremum of $(L-2)2^{-\beta(G)}$ over simple graphs of rank at least R containing every cycle length $3, \dots, L$. A cycle count gives $\Phi \leq 1$. Vertex identifications make realizers connected.

Theorem 4.1 (Near-halving). *For every sufficiently large integer R ,*

$$\Phi(2^{256R}) \leq \left(\frac{1}{2} + \frac{20}{\log^{(2)} R}\right) \Phi(R) + 2^{2-R}. \quad (4.1)$$

Proof. Consider a connected graph of rank $r \geq 2^{256R}$ with every cycle length $3, \dots, L$. If $L < 2^{2R} + 1$, then $(L-2)2^{-r}$ is below the error term. Otherwise, repeatedly add a cycle of the least missing length. Stop when the union W first has rank $t \geq 2R$.

At preceding rank $s < 2R$, the added length is at most $2^s + 2$. The improved bound is $\phi 2^s + 3$ when $s \geq R$, where $\phi = \Phi(R)$. Each step increases rank. Thus the process uses at most $2R$ cycles, and $\text{comp}(W) \leq 2R$. With $m = |E(W)|$, we obtain

$$m \leq 2^{2R} + 4R - 1 \leq 2^{3R}, \quad m + 1 \leq \phi 2^{2R} + 2^R + 5R, \quad (m+1)2^{-t} \leq \phi + 2^{1-R}.$$

Restriction to $M = E \setminus W$ has 2^{r-t} equally likely words. A mixed cycle requires an outside word that is a forest. Mixed cycles have at most $m+1$ lengths per outside word. Pure outside cycles contribute at most one length per word. Inside cycles contribute at most m lengths. Thus

$$(L-2)2^{-r} \leq (m+1)2^{-t} P_G(M) + 2^{-t} + m2^{-r}.$$

Theorem 1.2 applies because $m \leq \sqrt{r}$ and $\text{comp}(W) \leq 2R \leq (\log r)/100$. Since $\log^{(3)} r \geq \log^{(2)} R$, the coefficient of ϕ is at most the coefficient in (4.1). Also use $P_G(M) \leq 1$. The remaining terms satisfy

$$2^{1-R} + 2^{-2R} + 2^{3R-r} \leq 2^{2-R}$$

for large R . This proves (4.1). $\square$

Corollary 4.2 (Lower bound). *Every pancyclic graph on n vertices has at least $n + \log n + \log_* n - O(1)$ edges.*

Proof. Choose an integer R_0 large enough for Theorem 4.1. Iterate $R_{j+1} = 2^{256R_j}$. For $j \geq 2$, the identity $\log^{(2)} R_j = 8 + 256R_{j-2}$ gives

$$\sum_j \frac{1}{\log^{(2)} R_j} < \infty \quad \text{and} \quad \sum_j 2^{j-R_j} < \infty,$$

and (4.1) gives $\Phi(R_j) = O(2^{-j})$.

Also, $\log_* R_j = j + O(1)$. For the upper bound, compare with $U_{j+1} = 2^{U_j}$. Prove $R_j \leq \sqrt{U_j}$ by induction, with a large fixed U_0 as the initial value. The reverse bound follows from $R_{j+1} \geq 2^{R_j}$. Monotonicity then gives

$$\Phi(r) = O(2^{-\log_* r}).$$

A pancyclic n -vertex graph with $n+h$ edges has rank $r = h+1$. Thus $n-2 = O(2^{r-\log_* r})$ and $n \leq 2^{r+1}$. These bounds give $\log_* n \leq \log_* r + 2$. Therefore $h \geq \log n + \log_* n - O(1)$. $\square$

5 The matching construction

Proposition 5.1. *For every $n \geq 3$ there is a pancyclic graph on n vertices with at most $n + \lfloor \log n \rfloor + \log_* n + 1$ edges.*

Proof. For $3 \leq n \leq 6$, start with an n -cycle $v_0 \cdots v_{n-1}$. The construction includes the chord $v_0 v_2$ when $n \geq 4$. It also includes $v_0 v_3$ when $n = 6$. These graphs satisfy the proposition.

For $n \geq 7$, choose the largest k with $2^k + k + 1 \leq n$. On an n -cycle, choose an arc with successive segment lengths $2^i + 1$, $0 \leq i < k$. Its segment endpoints are $v_0, \dots, v_k$. Add the k segment shortcuts and $v_0 v_k$. Binary subset sums give the cycle lengths

$$[k + 1, 2^k + k] \quad \text{and} \quad [n - 2^k + 1, n].$$

Maximality of k gives $n \leq 2^{k+1} + k + 1$. Thus at most one intermediate length is missing. If this length is missing, one additional chord supplies it.

Start at $t = k$. While $t > 2$, add $v_0 v_j$ for $j = \lceil \log t \rceil < t$. Its shortcuts give $[j + 1, 2^j + j]$. Since $2^j \geq t$, this extends the interval of available lengths down to $j + 1$. Replace t by j .

This process uses $\log_* k - 1$ further chords. The construction adds at most

$$k + \log_* k + 1 \leq \lfloor \log n \rfloor + \log_* n + 1$$

edges in total. □

Corollary 4.2 and Proposition 5.1 prove Theorem 1.1.

A Lean correspondence

Module paths are relative to Erdos1016/ at commit a38913e. Declaration names below omit namespace prefixes. The formal proof uses the same argument. Explicit constants replace some O -terms. Auxiliary rounding and intermediate estimates have minor differences. The stated forest bound and final upper bound agree exactly.

Paper	Module	Declaration
Thm 1.1	Main.lean	mainTheorem
Lemma 2.1	CycleSpace/Graphical/ MultigraphCutCorrelation.lean, CycleSpace/Graphical/ DisjointRegionCuts.lean	multigraph_zeroCut_pair, exceptional_partner_card_le
Cor. 2.2	Probability/Moments/ SmallCutRegionBound.lean	smallCut_forest_bound
§3.1	Graph/Cubicization/ MarkedReduction.lean	exists_marked_subcubic_reduction
§3.2	Cycles/Selection/ ShortRegionPacking.lean, Cycles/ Selection/RetainedHighGirth.lean	retained_girth
§3.3	Nonbacktracking/Spectrum/ DeficitTrace.lean, Cycles/Counting/ LowDegreeTraceCycles.lean	normalized_even_trace_lower, cycleWordMass_ge_log_ratio
§3.4	Probability/Moments/ SeedPairCorrelation.lean	normalizedSelectedSeed_pair
§3.5	Cycles/Geometry/ ActualCycleLinkBound.lean	actual_cycle_links_le

Paper	Module	Declaration
§3.6	Probability/Moments/ SeedFamilyAvoidance.lean, Probability/Moments/ InducedCycleAvoidance.lean	second_moment_le, forest_le_half_add_twenty
Thm 1.2	Probability/Avoidance/ FewComponentForest.lean	fewComponentForestEstimate
Thm 4.1	Extremal/Witness/CapacityBudget. lean, Extremal/Recurrence/ WitnessReduction.lean	exists_budgeted_witness, recurrence_of_few_component_ forest_estimate
Cor. 4.2	Extremal/Recurrence/ LinearExponentialDecay.lean	recurrence_logStar_bound
Prop. 5.1	Extremal/UpperBound.lean	upper_bound_integer

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